

WTW158 CALCULUS
SEMESTER TEST 1

Date: March 2019 Marks: 50 Time $2\frac{1}{2}$ hours

MARKS FOR SECTIONS B & C	
	35

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FIRST NAMES

STUDENT NUMBER

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SIGNATURE

CONTACT NUMBER

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CIRCLE YOUR LECTURE GROUP NUMBER

1	2	3	4	5
Ms Yani	Prof Harding	Dr Garba	Ms Dreyer	Ms Mabitsela

READ THE FOLLOWING INSTRUCTIONS

1. The paper consists of pages 1 to 12. Check whether your paper is complete.
2. No calculator may be used.
3. No question paper may be removed from the venue.
4. Only write in black or blue pen.
5. No correction fluid (Tippex) allowed.

Section A:

Marks: 15 (1.5 per question)

1. The responses to questions in Section A must be completed on SIDE 2 of the optical reader form in SOFT PENCIL.
2. First circle your answers on this paper and then transfer these to the optical reader form.
3. There is only ONE correct answer per question.

Question 1

The solution set of $\frac{|x-2|}{x^2-1} > 0$ is

(a)	$(-\infty, 1) \cup (1, \infty)$	(b)	$(1, \infty) \cup (-1, \infty)$	☒ (c)	$(-\infty, -1) \cup (1, \infty) \setminus \{2\}$
(d)	$(-1, 1) \setminus \{2\}$	(e)	$(-1, 1)$	(f)	None of these

Question 2

The solution set of $|x| \leq 3$ and $|x| \geq 2$ is

(a)	$(-\infty, 3) \cup (3, \infty)$	☒ (b)	$[-3, -2] \cup [2, 3]$	(c)	$[-3, 3] \setminus [-2, 2]$
(d)	$[2, 3]$	(e)	$[-3, 3]$	(f)	None of these

Question 3

The domain of the function $f(x) = \sqrt{1 - \ln x}$ is

(a)	$(-\infty, e]$	☒ (b)	$(0, e]$	(c)	$(0, 1]$
(d)	(e, ∞)	(e)	$(1, \infty)$	(f)	None of these

Question 4

The range of the function $f(x) = |x| - x$ is

(a)	$(-\infty, 0]$	☒ (b)	$[0, \infty)$	(c)	$(-\infty, 1]$
(d)	$\mathbb{R}$	(e)	$\mathbb{R} \setminus \{0\}$	(f)	None of these

Question 5

If $\sec \theta = -2$ and $0 \leq \theta < 2\pi$, then $\theta =$

(a)	$\frac{2\pi}{3}$ or $\frac{4\pi}{3}$	(b)	$\frac{3\pi}{4}$ or $\frac{5\pi}{4}$	(c)	$\frac{7\pi}{6}$ or $\frac{11\pi}{6}$
(d)	$\frac{5\pi}{6}$ or $\frac{7\pi}{6}$	(e)	$\frac{4\pi}{3}$ or $\frac{5\pi}{3}$	(f)	None of these

Question 6

The value of $\operatorname{cosec}\left(-\frac{11\pi}{2}\right)$ is

(a)	-1	(b)	1	(c)	$\frac{\sqrt{3}}{2}$
(d)	$-\frac{\sqrt{3}}{2}$	(e)	$\frac{1}{2}$	(f)	None of these

Question 7

f is a function and a, b are real numbers. Consider the following statements:

- A. If $f(a) = f(b)$ then $a = b$
- B. If f is odd and $f(a) = -b$ then $f(-a) = b$
- C. If f is an even function then f^{-1} exists

Which of these statements is/are true?

(a)	Only A	(b)	Only B	(c)	Only C
(d)	A and B	(e)	A and C	(f)	B and C
(g)	All of these are true	(h)	None of these are true		

Question 8

The solution set of $\frac{2-x-x^2}{e^{x^2-1}} \geq 0$ is

(a)	$[-1, 2]$	(b)	$\mathbb{R}$	(c)	$\mathbb{R} \setminus (-2, 1)$
(d)	$[-2, 1]$	(e)	$[-2, 1)$	(f)	None of these

Question 9

Given the function $f(x) = \begin{cases} e^{x-3} & \text{if } x > 0 \\ -x-1 & \text{if } x < 0 \end{cases}$

$(f \circ f)(-4) =$

(a)	1	(b)	3	(c)	Undefined
(d)	0	(e)	$\ln 3$	(f)	None of these

Question 10

Given that a and b are any positive real numbers. Consider the following statements:

A. $\ln \frac{1}{a} = -\ln a$

B. $\frac{\ln a}{\ln b} = \ln(a-b)$

C. $\frac{e^a(e^{-b})^2}{\sqrt{e^{a+b}}} = e^{\frac{1}{2}(a-5b)}$

Which of these statements is/are true?

(a)	Only A	(b)	Only B	(c)	Only C
(d)	A and B	(e)	A and C	(f)	B and C
(g)	All of these are true	(h)	None of these are true		

Section B Marks: 13

1. This section must be completed IN PEN on this paper (including sketches).
2. Fill in the answer only in the space provided. Rough work can be done on the facing page and will not be marked.

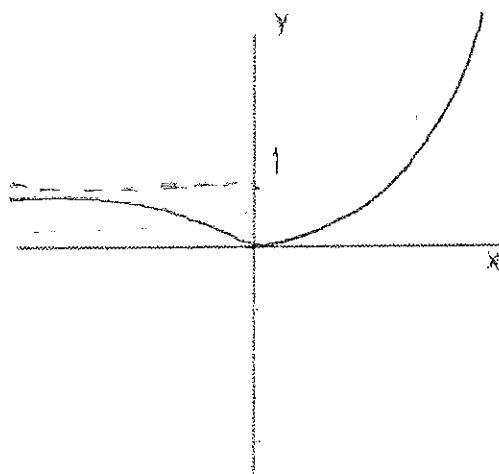
Question 11

11.1 Write the function $f(x) = |1 - e^x|$ as a piece-wise defined function.

$$f(x) = \begin{cases} 1 - e^x & \text{if } x \in (-\infty, 0] \\ -1 + e^x & \text{if } x \in (0, \infty) \end{cases}$$

2

11.2 Sketch the graph of $f(x) = |1 - e^x|$. Indicate all significant values.



2

Question 12

If $f(x) = \frac{1}{x-1}$ and $g(x) = |2-x|$, find $(f \circ g)(x)$ and determine the domain of $f \circ g$.

$$(f \circ g)(x) = \frac{1}{|2-x|-1}$$

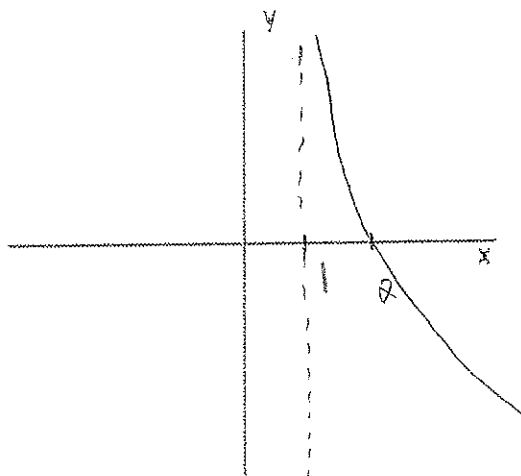
$$D_{f \circ g} = \mathbb{R} \setminus \{1, 3\}$$

$$\text{or } (-\infty, 1) \cup (1, 3) \cup (3, \infty)$$

2

Question 13

Sketch the graph of the function $f(x) = -\ln(x - 1)$ clearly indicating all significant values



2

Question 14

$$\arctan\left(\tan \frac{3\pi}{4}\right) = \tan^{-1}\left(\tan \frac{3\pi}{4}\right) =$$

$$-\frac{\pi}{4}$$

1

Question 15

15.1 Simplify by writing in terms of x only: $\cos(\arcsin x) = \cos(\sin^{-1} x) =$

$$\sqrt{1-x^2}$$

1

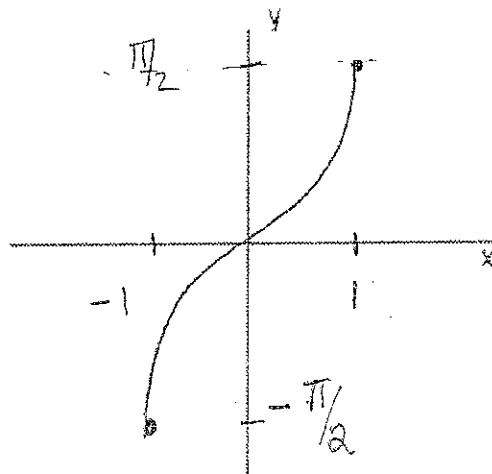
15.2 Simplify by writing in terms of x only: $\sin(2 \arcsin x) = \sin(2 \sin^{-1} x) =$

$$2x\sqrt{1-x^2}$$

1

Question 16

Sketch the graph of the function $f(x) = \arcsin x$ indicating all significant values.



2

Section C Marks: 22

1. The responses to questions in Section C must be completed IN PEN on this paper (including sketches).
2. All calculations must be shown and will be marked.
3. For more than the available space, use the facing page and clearly indicate it.
4. Attention should be paid to writing in a mathematically sound way.

Question 17

17.1 Show that the function $f(x) = \frac{3-x}{1-x}$ is its own inverse.

$$y = \frac{3-x}{1-x}$$

Inverse: $x = \frac{3-y}{1-y}$

$$\Rightarrow x(1-y) = 3-y$$

$$\Rightarrow -xy + y = 3-x$$

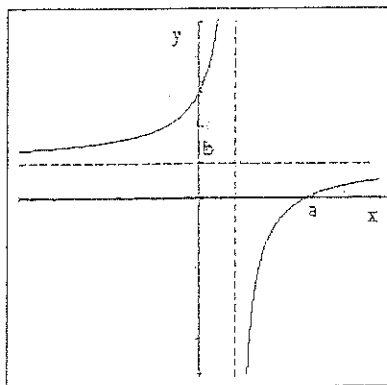
$$\Rightarrow y(1-x) = 3-x$$

$$\Rightarrow y = \frac{3-x}{1-x}$$

The function f in 17.1 is shown in the sketch. Complete the following:

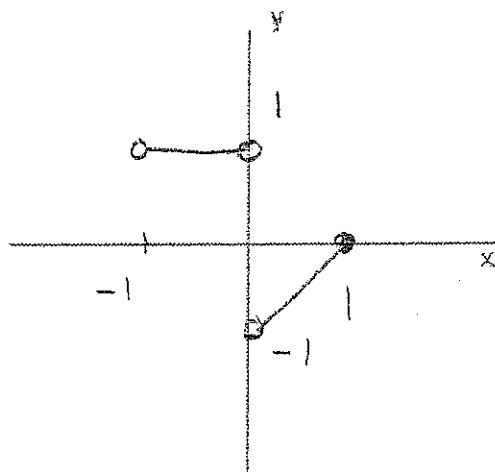
$a = \dots 3 \dots$

$b = \dots 1 \dots$



Question 18

18.1 Sketch the graph of the piece-wise defined function $f(x) = \begin{cases} 1 & \text{if } -1 < x < 0 \\ x-1 & \text{if } 0 < x \leq 1 \end{cases}$



18.2 Give the domain and range of f

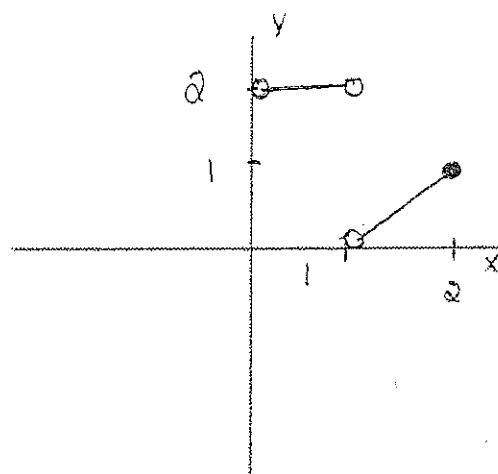
Domain: $(-1, 1] \setminus \{0\}$

Range: $(-1, 0] \cup \{1\}$

18.3 Give the piece-wise formulation of $g(x) = f(x-1) + 1$

$$g(x) = \begin{cases} 2 & \text{if } x \in (0, 1) \\ x-1 & \text{if } x \in (1, 2] \end{cases}$$

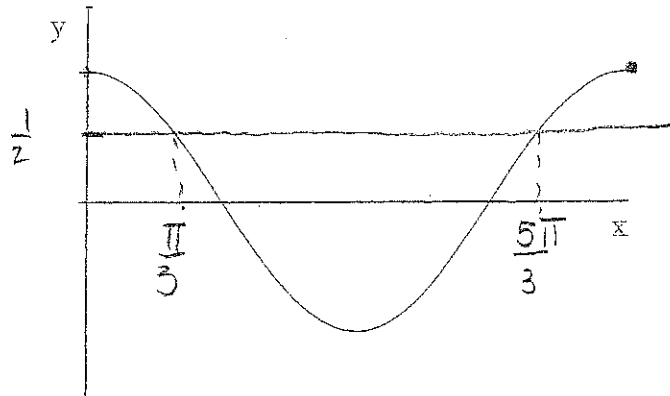
18.4 Sketch the graph of g



Question 19

The function $y = \cos x$ is given in the sketch on the interval $[0, 2\pi]$. Solve for x on $[0, 2\pi]$ such that $2 \cos x - 1 \geq 0$. Use the graph and indicate all applicable values on the sketch. Give the solution in interval notation.

$$\cos x \geq \frac{1}{2}$$



Solution: $x \in [0, \frac{\pi}{3}] \cup [\frac{5\pi}{3}, 2\pi]$

3

Question 20

Solve for x if $2 \cos^2 x + \sin x = 1$

$$2(1 - \sin^2 x) + \sin x = 1$$

$$\Rightarrow -2 \sin^2 x + \sin x + 1 = 0$$

$$\Rightarrow (-2 \sin x - 1)(\sin x - 1) = 0$$

$$\Rightarrow \sin x = -\frac{1}{2} \quad \text{or} \quad \sin x = 1$$

$$\Rightarrow x = \frac{7\pi}{6} + 2k\pi \quad \text{or} \quad x = \frac{11\pi}{6} + 2k\pi \quad \text{or} \quad x = \frac{\pi}{2} + 2k\pi$$

$$\text{or} \quad x = -\frac{\pi}{6} + 2k\pi \quad \text{or} \quad x = -\frac{5\pi}{6} + 2k\pi \quad \text{or} \quad x = \frac{\pi}{2} + 2k\pi$$

Question 21

Solve for x : $\ln\left(\frac{1}{x}\right) + \ln(2x^3) = \ln 4$.

$$\Rightarrow \ln\left(\frac{2x^3}{x}\right) = \ln 4$$

$$\Rightarrow \ln 2x^2 = \ln 4$$

$$\Rightarrow 2x^2 = 4$$

$$\Rightarrow x^2 = 2$$

$$\Rightarrow x = \sqrt{2} \quad (x \neq -\sqrt{2})$$

$$\begin{aligned} \ln 1 &= \ln x + \ln 2 + 3 \ln x = \ln 4 \\ \Rightarrow 2 \ln x &= \ln 2 \\ \Rightarrow x &= \sqrt{2} \end{aligned}$$

Question 22

Determine whether the function $f(x) = \frac{1+e^x}{1-e^x}$ is even/odd or neither.

$$f(-x) = \frac{1+e^{-x}}{1-e^{-x}}$$

$$= \frac{1+\frac{1}{e^x}}{1-\frac{1}{e^x}}$$

$$= \frac{\frac{e^x+1}{e^x}}{\frac{e^x-1}{e^x}}$$

$$= \frac{e^x+1}{e^x-1}$$

$$= -\left(\frac{e^x+1}{1-e^x}\right) = -f(x)$$

$\Rightarrow f$ is odd.

Question 23

Solve for x if $\frac{1}{\ln x} > 1$. Show all reasoning and calculations.

$$\Rightarrow \frac{1}{\ln x} \times (\ln x)^2 > (\ln x)^2$$

$$\Rightarrow \ln x - (\ln x)^2 > 0$$

Let $\ln x = k$

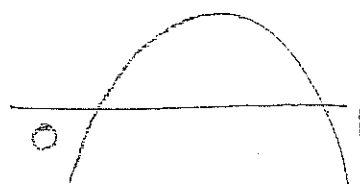
$$\Rightarrow k - k^2 > 0$$

$$\Rightarrow k(1-k) > 0$$

$$\Rightarrow 0 < k < 1$$

$$\Rightarrow 0 < \ln x < 1$$

$$\Rightarrow \underline{x \in (1, e)}$$



3

(a) $\frac{1}{\ln x} \cdot (\ln x)^2 > (\ln x)^2$

$$\Rightarrow \ln x - (\ln x)^2 > 0$$

$$\Rightarrow \ln x(1 - \ln x) > 0$$

$$\Rightarrow \ln x > 0 \text{ and } 1 - \ln x > 0 \text{ or } \ln x < 0 \text{ and } 1 - \ln x < 0$$

$$\Rightarrow x > 1 \text{ or } x < e \text{ or } x \in \emptyset$$

$$\Rightarrow x \in (1, e)$$